Adaptive SoH Forecasting for Second-Life LFP Cells Using PyBaMM-Generated Data and Hybrid Model Selection

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Second-life reuse of retired lithium iron phosphate cells requires rapid prediction of future degradation for each incoming cell. Rather than assuming that one forecasting model is optimal for all cells, we combine exponential extrapolation with a PyBaMM [1]-trained neural operator and select between them using early-cycle behaviour.

A short diagnostic and the first $K=50$ operating cycles provide the model context. A Neural ODE [3] was trained on 490 PyBaMM SPMe degradation trajectories generated around seven anchor cells from three LFP suppliers (CALB, EVE, REPT) on Prada2013 chemistry [2], using trajectory-grouped train/validation partitions. In parallel, an exponential model was fitted to the same context; its first-80%-fit, last-20%-prediction context-holdout residual indicates departure from exponential behaviour. A threshold set on synthetic trajectories alone routes each cell to either forecast (Figure 1).

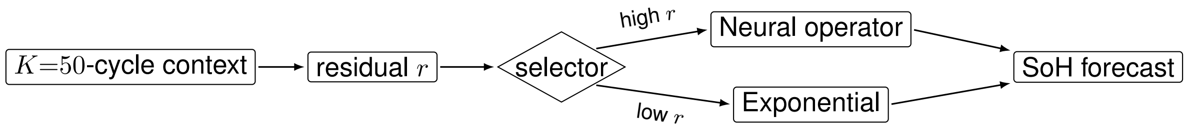

Figure 1: Adaptive selector. A context-holdout residual r (exp fit on the first 80% of the $K=50$ context, RMSE on the last 20%) routes each cell to either the PyBaMM-trained neural operator (high r) or exponential extrapolation (low r); the threshold is set on synthetic trajectories only.

Across three external cells excluded from operator training, exponential and neural forecasts achieved mean SoH RMSEs of 0.70 and 0.80 percentage points, respectively (Figure 2). Across these three cells, the frozen residual-based selector chose the lower-error model for all three and reduced mean RMSE to **0.47 percentage points**, a 33% improvement relative to the best fixed strategy. Exponential extrapolation performed best for CALB_0029 and REPT_0028, whereas the neural operator performed best for EVE_0003.

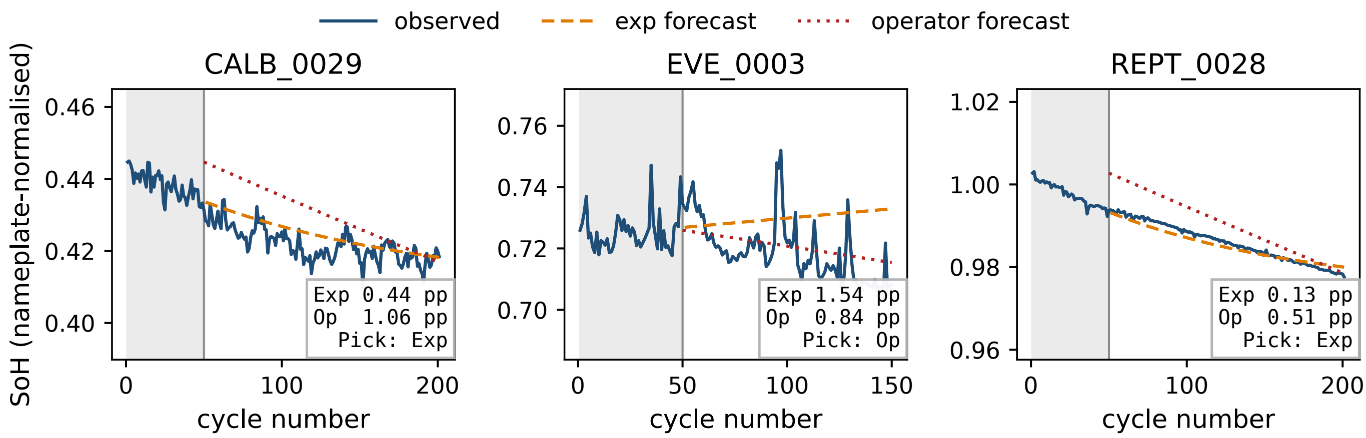

Figure 2: Forecasts evaluated against measured post-context SoH for three external cells. Grey shading denotes the $K=50$ context window; orange dashed and red dotted lines show exponential and neural-operator forecasts, respectively. Panels use independent y-axis ranges.

Matched ablations showed little additional benefit from explicit degradation-parameter conditioning: full-, no- and reduced-θ operators achieved mean RMSEs of 0.80, 0.82 and 0.90 percentage points, respectively. These results suggest that matching model complexity to observed early-cycle behaviour may be more effective than applying one forecasting model uniformly across heterogeneous second-life cells. Evaluation is currently limited to three external cells, and longer-horizon projections require additional experimental validation.

[1] V. Sulzer et al., *J. Open Res. Software* **9** (2021), 14. • [2] M. Prada et al., *J. Power Sources* **219** (2012), 353–362. • [3] R. T. Q. Chen et al., *NeurIPS* (2018).